

Hotel & Airbnb Booking Analytics System — Portfolio Project

Business Scenario

A rapidly growing hospitality company operating hotels and Airbnb-style rental properties across multiple cities is facing challenges in managing and analyzing its business performance.

The company handles thousands of bookings every month through multiple booking platforms and channels. Over time, management has noticed inconsistencies in revenue growth, increasing booking cancellations, fluctuating occupancy rates, changing customer behavior, and varying service quality across properties.

Currently, business reports are manually created using spreadsheets, making it difficult for management to track performance, identify operational issues, monitor customer satisfaction, and make data-driven decisions efficiently.

The company wants to implement a centralized Business Intelligence system using Microsoft Power BI that can provide real-time insights into:

- Revenue & profitability
- Booking trends
- Guest behavior
- Occupancy performance
- Cancellation trends
- Property performance
- Customer satisfaction
- Regional business growth

The final solution should help management improve operational efficiency, increase profitability, optimize pricing strategies, improve guest experience, and support strategic decision-making.

Business Objective

As a Business Intelligence Developer/Data Analyst, your task is to design and develop a complete Hotel & Airbnb Booking Analytics System using Microsoft Power BI.

The business is looking for a solution that can:

- Monitor overall business performance
- Identify high-performing and underperforming properties
- Analyze booking and revenue trends over time
- Understand guest behavior and loyalty patterns

- Track occupancy and cancellation rates
- Improve customer satisfaction through operational insights
- Support management with interactive and real-time reporting
- Enable better business decision-making using data

The goal is to transform raw booking data into actionable business insights through professional dashboards, data modeling, and analytical storytelling.

Project Overview

This project focuses on building a complete Hotel & Airbnb Booking Analytics System using Microsoft Power BI.

Students are expected to perform end-to-end Business Intelligence development, including:

- Data Cleaning & Transformation (ETL)
- Star Schema Data Modeling
- Advanced DAX Calculations
- Time Intelligence Analysis
- Interactive Dashboard Development
- Business Insight Generation
- Professional Reporting & Storytelling

The objective of this project is to simulate a real-world hospitality analytics environment where hotel managers, business owners, operations teams, and decision-makers can monitor booking performance, customer behavior, operational efficiency, profitability, and guest satisfaction through interactive dashboards.

This project is designed as a portfolio-grade analytics solution that students can showcase in:

- Job interviews
- LinkedIn portfolios
- Freelance portfolios
- Data Analyst / BI Developer resumes
- Final year projects

Business Intelligence System Approach

Students should treat this project as a complete Business Intelligence System consisting of:

1. Data Source Layer
2. Data Processing Layer (ETL)
3. Data Modeling Layer
4. Business Logic Layer (DAX)

5. Presentation & Reporting Layer

Dataset Information

Dataset Name

HotelAirbnb_PortfolioProject.csv

Dataset Description

The dataset contains approximately 150,000 hotel and Airbnb booking records.

Each row represents one guest booking transaction.

The dataset includes information related to:

- Booking Details
- Hotel & Airbnb Properties
- Guest Information
- Room Details
- Revenue & Pricing
- Discounts & Taxes
- Customer Reviews & Ratings
- Loyalty Programs
- Complaints & Service Quality
- Operational Information

Data Cleaning & Transformation (ETL)

Students must perform all required data preparation steps using Power Query.

Required Tasks

Data Validation

- Verify and correct all data types
- Standardize inconsistent date formats
- Validate revenue and profit calculations
- Check for invalid or inconsistent values

Missing Values Handling

- Handle missing values in:

- Meal_Plan
- Review_Given
- Cancellation_Reason
- Complaint_Category
- Special_Request

Duplicate Handling

- Check duplicate `Booking_ID`
- Remove duplicate records if necessary

Feature Engineering

Create the following calculated/derived columns:

- `Guest_Age_Group`
- `Loyalty_Status`
- `Booking_Lead_Category`
- `Revenue_Category`
- `Occupancy_Status`
- `Stay_Duration_Category`

Operational Handling

- Handle cancelled bookings properly
- Exclude cancelled bookings from occupancy calculations
- Validate room revenue calculations

Data Quality Checks

- Identify invalid ratings
- Check negative or unrealistic revenue values
- Validate booking and stay duration consistency
- Identify anomalies in guest records

Documentation Requirement

Students must clearly document all Power Query steps performed during ETL.

Data Modeling

Students are required to build a proper Star Schema model.

Fact Table

Fact_Bookings

Suggested Columns

- | | | | | |
|------------------|-----------------|---------------|----------------|---------|
| • Booking_ID | Guest_ID | Property_ID | Room_ID | Date_ID |
| • Payment_ID | Loyalty_ID | Check_In_Date | Check_Out_Date | |
| • Nights_Stayed | Total_Paid | Room_Revenue | Extras_Revenue | |
| • Discounts | Taxes | Gross_Profit | Guest_Rating | |
| • Booking_Status | Complaint_Count | | | |

Dimension Tables

Dim_Date

Suggested Columns:

- | | | | | | | |
|-----------|-----------|-----|-------|------------|---------|------|
| • Date_ID | Full_Date | Day | Month | Month_Name | Quarter | Year |
| • Weekday | | | | | | |

Dim_Guest

Suggested Columns:

- | | | | | |
|---------------|----------------|--------|-----|-----------------|
| • Guest_ID | Guest_Name | Gender | Age | Guest_Age_Group |
| • Nationality | Loyalty_Status | | | |

Dim_Property

Suggested Columns:

- | | | | | |
|---------------|-----------------|---------------|------|--------|
| • Property_ID | Property_Name | Property_Type | City | Region |
| • Country | Rating_Category | | | |

Dim_Room

Suggested Columns:

- | | | | |
|-----------|-----------|---------------|-----------|
| • Room_ID | Room_Type | Room_Capacity | Meal_Plan |
|-----------|-----------|---------------|-----------|

Dim_Location

Suggested Columns:

- | | | | | |
|---------------|------|-------|---------|--------|
| • Location_ID | City | State | Country | Region |
|---------------|------|-------|---------|--------|

Dim_Payment

Suggested Columns:

- Payment_ID Payment_Method Currency

Dim_Loyalty

Suggested Columns:

- Loyalty_ID Membership_Tier Loyalty_Status Reward_Points

Dim_Booking

Suggested Columns:

- Booking_Source Booking_Channel Booking_Lead_Category
- Cancellation_Reason

Requirements

- Create proper relationships between tables
- Use 1-to-many relationships where applicable
- Create a dedicated _Measures table
- Create and mark a Date table properly
- Maintain a clean and optimized model

DAX Measures & Calculations

Students must create advanced DAX measures.

Core Measures

Revenue & Sales

- Total Revenue
- Total Bookings
- Average Booking Value
- Extras Revenue
- Revenue per Guest

Profitability

- Total Gross Profit

- Profit Margin %
- Revenue Growth %

Operational Metrics

- Occupancy Rate
- Cancellation Rate
- Average Nights Stayed

Customer Metrics

- Average Guest Rating
- Loyalty Revenue Contribution
- Repeat Guest Rate
- Complaint Rate

Time Intelligence (Mandatory)

Students must implement Time Intelligence calculations using DAX.

Required Measures

- Revenue YTD
- Revenue MTD
- Revenue YoY %
- Month-over-Month Growth %
- Running Total Revenue
- Rolling 3-Month Revenue
- Quarterly Revenue Trends
- Previous Month Revenue Comparison

Dashboard Development

Students must develop a professional and interactive 5-page Power BI dashboard.

The dashboard should focus on:

- Business Performance Monitoring
- Revenue Analysis
- Guest Insights
- Operational Efficiency
- Property Performance

The design should be clean, professional, and business-oriented.

Page 1 — Executive Overview Dashboard

Purpose

This page provides a high-level overview of hotel and Airbnb business performance.

It should help decision-makers quickly understand revenue trends, booking activity, occupancy, guest satisfaction, and overall operational performance.

Required Visuals (Maximum 6 Visuals)

- KPI Cards
- Monthly Revenue Trend
- Revenue by Property Type
- Revenue by City
- Booking Status Breakdown
- Top Properties by Revenue

KPI Requirements

- Total Revenue
- Total Bookings
- Average Booking Value
- Average Guest Rating
- Occupancy Rate

Business Questions

Students must answer:

- What is the overall business performance?
- Is revenue increasing or decreasing month-over-month?
- Which property type generates the highest revenue?
- Which cities generate the most bookings?
- Which months generate the highest sales?
- How does occupancy rate change over time?
- What are the Year-to-Date (YTD) revenue trends?
- How does current month performance compare with last month?

Page 2 — Sales & Revenue Analysis

Purpose

This page focuses on detailed sales performance and profitability analysis.

It should help identify revenue drivers, profitable properties, booking trends, and pricing behavior.

Required Visuals (Maximum 6 Visuals)

- Revenue by Property Type
- Revenue by Room Type
- Revenue by Booking Source
- Profit Margin Analysis
- Revenue vs Nights Stayed Scatter Plot
- Running Revenue Trend

Business Questions

Students must answer:

- Which properties are most profitable?
- How do discounts affect overall revenue?
- Which payment method is most commonly used?
- Which booking source generates the highest revenue?
- Do longer stays generate more revenue?
- Which quarter generated the highest revenue?
- Is revenue growing consistently over time?
- How does this month's revenue compare with previous months?
- Which periods experience the highest booking performance?

Page 3 — Guest & Loyalty Analysis

Purpose

This page focuses on guest segmentation, loyalty behavior, and booking trends.

It should help understand guest preferences and identify high-value customer groups.

Required Visuals (Maximum 6 Visuals)

- Guest Age Distribution
- Revenue by Age Group & Room Type
- Loyalty vs Non-Loyalty Analysis
- Repeat Guest Analysis

- Booking Source Trends
- Guest Rating Analysis

Business Questions

Students must answer:

- Which guest segment spends the most?
 - Do loyalty members generate more revenue?
 - Which booking channels are growing fastest?
 - Which age group prefers premium rooms?
 - Are loyalty memberships increasing over time?
 - Which months have the highest guest bookings?
 - Which customer groups generate repeat business?
 - Which property types receive the highest ratings?
-

Page 4 — Operational Performance Dashboard

Purpose

This page focuses on operational efficiency and service quality.

It should help identify cancellation trends, complaint patterns, and operational issues affecting guest satisfaction.

Required Visuals (Maximum 6 Visuals)

- Booking Status Breakdown
- Cancellation Analysis
- Complaint Category Analysis
- Rating Distribution
- Guest Satisfaction Trends
- Property Performance Comparison

Business Questions

Students must answer:

- Which properties experience the highest cancellations?
- Which complaint categories occur most frequently?
- Do complaints reduce guest ratings?

- Which cities face the most operational issues?
- Are cancellations increasing over time?
- Which months experience the highest operational issues?
- Which properties maintain the best guest satisfaction?
- Which staff members handle the most bookings?

Page 5 — Property & Regional Performance Analysis

Purpose

This page focuses on regional performance, property profitability, and demand distribution.

It should help evaluate booking demand and business performance across cities and regions.

Required Visuals (Maximum 6 Visuals)

- Top Properties by Revenue
- Top Cities by Bookings
- Revenue by Region
- Revenue by Nights Stayed
- Regional Performance Map
- Revenue by Property Trend

Business Questions

Students must answer:

- Which properties generate the highest revenue?
- Which cities generate the most guest traffic?
- Are property revenues increasing over time?
- Which months generate the highest booking demand?
- Which regions contribute most to total revenue?
- Which properties have the best operational performance?
- How does stay duration impact profitability?
- Which property type performs best overall?

Slicers & Interactivity

Students must implement a consistent slicer panel across all dashboard pages.

Required Slicers

- Date Range
- Property Type
- Property City
- Booking Status
- Room Type
- Loyalty Status
- Booking Source
- Payment Method

Interactivity Requirements

- Sync slicers across pages
- Drill-through functionality
- Report tooltips
- Bookmarks for alternate views
- Dynamic titles
- Conditional formatting

Portfolio Report Documentation

Students must submit a professional PDF report.

Report Structure

1. Dashboard Overview

Explain:

- Purpose of the dashboard
- Business problem solved
- Key outcomes and insights

2. ETL Process

Explain:

- Data cleaning steps
- Missing value handling
- Feature engineering
- Validation process

3. Data Modeling

Explain:

- Fact and dimension tables
- Relationships
- Star Schema design

4. Dashboard Pages

For each page:

- Explain purpose
- Answer business questions
- Highlight key findings

5. Time Intelligence Analysis

Explain:

- YTD
- MTD
- MoM
- YoY
- Rolling trends

6. Data Quality Findings

Mention:

- Missing values
- Duplicate records
- Invalid entries
- Data inconsistencies

7. Conclusion

Summarize major findings and business impact.

8. Recommendations

Provide actionable business recommendations based on analysis.

9. Self Reflection (Optional)

Students may include:

- Challenges faced
- Skills learned
- Improvements for future work

Deliverables

1. Power BI File (.pbix)

Must include:

- 5 dashboard pages
- Star Schema model
- DAX measures
- Time Intelligence calculations
- Interactive slicers and filters
- Professional dashboard design

2. PDF Portfolio Report

Must include:

- Dashboard overview
- ETL documentation
- Data model explanation
- Insights and recommendations
- Conclusion

Bonus Features (Optional)

Students may implement advanced features for additional portfolio impact:

- Row-Level Security (RLS)
- Advanced Tooltips
- Hidden Navigation Pages
- Custom Themes

Best Practices

- Keep dashboards clean and professional
- Use consistent color themes

- Avoid overcrowding visuals
- Use meaningful titles and labels
- Validate calculations before reporting
- Focus on storytelling with data
- Use KPIs to highlight performance trends
- Ensure dashboard usability and readability

Final Objective

The goal of this project is to build a professional hospitality analytics solution that demonstrates:

- Power BI expertise
- ETL & data modeling skills
- Advanced DAX capabilities
- Business intelligence thinking
- Analytical storytelling
- Dashboard design skills
- Real-world portfolio readiness